



Tribhuvan University

Faculty of Humanities and Social Sciences

Dog Adoption System

A PROJECT REPORT

Submitted To:

Department of Computer Application

Mega National College

*In partial fulfillment of the requirements for the Bachelor's degree in Computer
Application*

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by Binita Shrestha and Prashna Rai entitled "Dog Adoption System" in partial fulfillment of the requirement for the degree of Bachelor of Computer Application is recommended for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by BINITA SHRESTHA and PRASHNA RAI entitled "Dog Adoption System" in partial fulfillment of the requirement for the degree of Bachelor's in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as project for the required degree.

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ABSTRACT

The Dog Adoption System (DAS) project is designed to make the process of adopting dogs easier, faster, and more organized. It works as an online platform where all adoption-related tasks can be managed in one place from creating dog profiles and receiving adoption applications to approving requests and keeping post-adoption records.

With DAS, shelter staff i.e admin can add detailed information for each dog, including breed, age, health status, vaccination dates, and behaviour notes. They can also upload photos , update adoption availability, review applications for adoption. The system can also keep track of important records such as vaccination reminders, medical check-ups and other necessities.

For adopters, DAS allows them to browse available dogs, search by breed, age, size, or temperament, submit adoption applications online, track their application status, and receive updates from the shelter. It also provides announcements for upcoming adoption events and special programs.

By bringing all these features together, the Dog Adoption System improves the adoption process, makes it more transparent, reduces manual work for staff, and helps match the right dog with the right adopter. Authorized staff can log in anytime to see updated records, ensuring smooth and efficient adoption management.

Keywords : Dog Adoption System, Pet Adoption, Dog Profiles, Adoption Management, Shelter Records.

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LIST OF ABBREVIATIONS

CPU	Central Processing Unit
CSS	Cascading Style Sheet
DAS	Dog Adoption System
DFD	Data Flow Diagram
ER	Entity Relationship
GB	Giga Byte
iOS	Iphone Operating System
JS	Javascript
MySQL	My Structured Query Language
PHP	Hypertext Preprocessor
SMS	Short Message Service
RAM	Random Access Memory
UI/UX	User Interface and User Experience
XAMPP	Cross-platform, Apache, MariaDB (formerly MySQL), PHP, and Perl

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Chapter 1: INTRODUCTION

1.1 Introduction

The Dog Adoption Project aims to create a user-friendly online platform that simplifies the pet adoption process for individuals looking to provide a loving home to dogs in need. This project seeks to bridge the gap between dog shelters and potential adopters through a digital solution. The platform will allow users to register, browse adoptable dogs and fill out adoption forms. The primary goal of this project is to increase the adoption rate of dogs by making the process more accessible, transparent, and efficient. Through the platform, users can easily find detailed information about available dogs. Additionally, by digitizing the system and approval workflow, shelters can manage requests more effectively and reduce administrative overhead. Ultimately, the project contributes to the broader mission of reducing the number of homeless dogs and encouraging compassionate, lifelong pet guardianship.

1.2 Problem Statement

Despite the growing number of dog shelters and rescue centers, many dogs remain unadopted due to the lack of an efficient, accessible, and user-friendly platform that connects them with potential adopters. Traditional adoption processes are often time-consuming, confusing, and limited to physical visits, which discourages many willing individuals from adopting. Furthermore, shelters struggle with managing adoption applications manually, leading to delays and missed opportunities. This project addresses these challenges by developing an online system that streamlines the dog adoption process, making it easier for both shelters and adopters to connect, communicate, and complete adoptions successfully.

1.3 Objective

1. To develop an easy-to-use online platform where users can create accounts and securely apply for dog adoption.
2. To simplify the dog adoption process by allowing users to fill out and submit adoption forms digitally.
3. To increase the chances of dogs finding suitable and loving homes by streamlining the adoption workflow.

1.4 Scope and Limitations

The dog adoption system is designed to provide an online platform where potential adopters can create accounts, browse available dogs, and submit adoption applications. It allows shelter administrators to add, update, or remove dog listings, review applications, and approve or reject adoption requests. The system aims to simplify the adoption process by offering a centralized database of adoptable dogs with details such as breed, age, and health status. It also enhances communication between shelters and adopters through updates and notifications, ensuring a smooth and transparent process.

While the system offers many useful features, it has certain limitations. It does not currently include a payment gateway for adoption fees or donations. Real-time chat or video interaction between adopters and shelter staff is not yet implemented. The system relies on the accuracy of information entered by the shelter, meaning errors or outdated details could affect the adoption process. Additionally, it is dependent on internet access, which may limit accessibility for users in remote areas. The platform also requires regular updates and maintenance to remain functional and secure.

1.5 Development Methodology

The incremental model is a software development methodology in which the system is designed, implemented, and tested incrementally until the full product is complete. The first increment usually contains the core functionalities, and each subsequent increment adds new features or enhances existing ones. This allows for early delivery of working software, making it easier to identify and fix issues at an early stage. It also enables regular feedback from users, helping developers adjust to changing requirements more effectively. The incremental model is particularly useful for projects where requirements are expected to evolve over time.

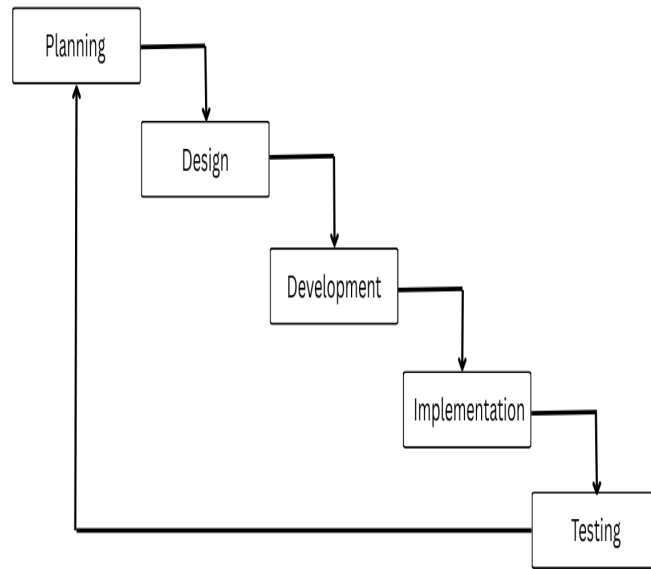


Figure 1.1: Incremental Model

1.6 Report Organization

This report is organized into the following major sections to provide a comprehensive overview of the Dog Adoption System, from conceptual foundations to implementation and evaluation:

- **Background Study and Literature Review**

- Overview of existing dog adoption systems

Introduces traditional and online dog adoption platforms, highlighting their core mechanisms and processes.

- Related research and technologies

Summarizes academic studies and technologies relevant to pet adoption, including database-driven web systems, search filters, and application tracking.

- Comparative analysis of similar platforms

Evaluates existing dog adoption solutions to identify strengths, limitations, and opportunities for improvement in the proposed system.

- **System Analysis and Design**

- **System Analysis**

- Requirement Analysis

- Details functional and non-functional requirements gathered from potential adopters, shelter staff, and administrators.

- Feasibility Study

- Assesses technical, economic, and operational feasibility of the proposed adoption system.

- Process Modeling (DFD)

- Uses Data Flow Diagrams to visualize adoption processes, from listing dogs to approving applications.

- **System Design**

- Architectural Design

- Database Schema Design

- Key Entities

- Interface Design (UI/UX)

- Illustrates user interface layouts and usability considerations for adopters, shelter staff, and administrators.

- Physical DFD

- Provides a detailed physical-level Data Flow Diagram showing actual data stores, external entities, and system components.

- **Implementation and Testing**

- **Implementation**

- Tools Used

- Lists programming languages, and tools used to develop the system.

- Implementation Details

- User Registration and Authentication

- Handles secure login, registration, and role-based access for adopters and administrators.

- Dog Listing Module

- Allows administrators to add, update, and delete adoptable dog profiles with images and details.

- Adoption Application Module

- Enables users to submit adoption requests and track application status.

- **Admin Panel Module**
Provides administrative controls for managing dog listings, applications, and user accounts.
- **Database Integration**
Explains how the system communicates between the user interface and the database.
- **Testing**
 - **Test Cases for Unit Testing**
Validates individual components such as registration, search filters, and application submission.
 - **Test Cases for System Testing**
Ensures the entire system functions as intended under different usage scenarios.
- **Conclusion and Future Recommendations**
 - **Conclusion**
Summarizes the objectives, implementation, and overall performance of the Dog Adoption System.
 - **Lessons Learned**
Reflects on challenges faced during development and improvements in design and usability.
 - **Future Recommendations**
Suggests enhancements such as mobile app integration, real-time chat between adopters and shelters, and AI-based dog matching algorithms.

Chapter 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Dog adoption is a widely recognized method of giving stray, abandoned, or surrendered dogs a second chance at life in a caring home. Across the world, countless dogs end up in shelters due to various reasons—owner abandonment, abuse, breeding overpopulation, or natural disasters. While shelters serve as temporary safe havens, their limited capacity and resources often lead to overcrowding, stress, and even euthanasia in extreme cases.

Over the years, animal welfare organizations and governments have launched numerous campaigns to promote dog adoption, resulting in increased public awareness in some regions. However, in many communities—especially in developing areas—the adoption rate remains low due to a lack of education, stigma around shelter animals, and insufficient adoption infrastructure. Studies have shown that adopted dogs can form equally strong bonds with their new families as purchased pets. Moreover, adoption helps break the cycle of puppy mills and unethical breeding practices, contributing to more humane treatment of animals overall.

2.2 Literature Review

The issue of pet overpopulation, particularly among dogs, has prompted increased focus on adoption as a humane and sustainable solution. Research across various disciplines, including veterinary science, animal behavior, sociology, and information technology, provides insights into the complex dynamics of dog adoption and the factors influencing adoption outcomes.

According to the statistics published by "The Kathmandu Post", existing data suggests a significant stray dog population, particularly in Kathmandu Valley, with estimates ranging from tens of thousands to over 100,000. These stray dogs are often referred to as "community dogs" as they are part of local communities and are common sight on almost every street in Kathmandu Valley. According to Sneha's Care, more than 10,000 street animals die every year in Nepal from illnesses, diseases and accidents[1]. Controlling stray dogs is not an easy job. Dogs without owners are difficult to tame and care for. At present, the Agriculture and Animal Husbandry Department has coordinated with nine different organizations to sterilize and vaccinate stray dogs in all the wards of the metropolitan. They include Sneha Care, Animal Nepal, Vet 4 Ur Pet, Kathmandu Animal Treatment Centre Nepal and others[2]. The lucky ones are unofficially 'adopted' by a local shop or kind-hearted individual. These 'community dogs' are at least regularly provided with food, albeit lacking in

protein and nutrition[3].

It's incredibly rewarding to hear about their rescue and adoption using technology and see them find their forever families. Helping rescue animals is truly a fulfilling experience[4]. Technology increases the visibility of dogs in need, giving countless animals a second chance at life in loving homes. With proper utilization of available technology, dog rescue and adoption can rise bridging the gap between homeless animals and people willing to offer them a second chance. Not only dog animal rescue is a much-needed social service that helps protect animals from dangerous situations that often involve cruelty and abuse[5]. Technology has become an essential tool in modern dog rescue and adoption efforts, bridging the gap between homeless animals and people willing to offer them a second chance. Dog rescue and adoption efforts benefit communities by reducing stray populations, decreasing dog bite incidents, and promoting humane attitudes toward animals.

When you adopt a rescue dog, you offer them a chance at a brighter future. Many rescue dogs come from difficult or uncertain backgrounds, often marked by hardship[6]. If you buy a puppy from a pet shop or a breeder, the chances are they come from a puppy mill where they are kept in terrible conditions. To people and shops selling puppies, profit and not the welfare of the dog is of utmost importance. By adopting a rescue, you can help lower the demand for puppies from puppy mills. The good news is that more Nepali families are now inclined towards keeping dogs and letting their children grow up with pets, instilling in them virtues of kindness and compassion towards all living beings from a young age[7]. Pets — from dogs and cats to birds, reptiles, and other animals — bring joy to our lives and warmth to our homes. Across the U.S., millions of animals in shelters are waiting for the chance to find a loving forever home. National Adopt a Shelter Pet Day, observed on April 30, aims to raise awareness and encourage adoptions to help them find families[8]. Rescue pets may come from tough beginnings, but with a second chance, they often become the most loyal companions. Not only dogs adopting animals gives a second chance at the life they deserve. When you adopt, you're not only changing their life, but they often end up changing yours too[9].

Adopting from an animal shelter offers a deserving pet a second chance while supporting the well-being of animals in need. Beyond the satisfaction of providing a home, the joy and unique bond formed between a shelter pet and its adopter are truly priceless[10]. Thankfully, by welcoming them into a safe and loving home, we can help them heal and experience happiness despite their past. Families who adopt not only gain a loyal companion but also help reduce the number of homeless animals.

Chapter 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

3.1.2 Functional Requirement

1. Enable users (such as adopters, volunteers, and donors) to create personal accounts.
2. Provide a platform where users can view all dogs available for adoption.
3. Allow users to manage their profiles, including updating information like name, contact details
4. Keep a detailed database of dogs available for adoption, including information like breed, age, health condition, and vaccination records according to the availability of the received information.

3.1.3 Non-Functional Requirement

The Dog Adoption System is designed to offer an easy and efficient platform for people to adopt dogs. Besides offering core functionalities like viewing, searching, and applying for adoption, the system ensures a smooth, secure, and reliable experience for users

3.2 Feasibility Study

1. Technical

Dog Adoption will be developed using open source modern technologies. The front-end will be built using JavaScript (JS) for dynamic page and PHP , XAMPP will be used for backend. XAMPP will be used as the database for reliable and efficient data storage.

Hardware Requirements

- (a) Intel Core i5 or higher CPU
- (b) Minimum 8 GB RAM
- (c) Minimum 100 GB storage

Software Requirements

- (a) Any modern web browser (Chrome, Edge, Firefox)
- (b) HTML, CSS, JavaScript
- (c) PHP, XAMPP

2. Operational

Many dogs are abandoned, stray, or surrendered and need homes. The rescue and adoption system would involve rescuing, medical care, fostering, and matching dogs with adopters. The system will be easy for users like dog rescuers, shelters, and adopters to operate within and offers a user-friendly interface with effective in daily rescue and adoption activities.

3. Economic

This project is economically feasible as it requires minimal investment compared to the potential service it offers. Development costs are low as free tools are used. JavaScript, PHP and MYSQL can be used which are open source.

2. Scheduling



Figure 3.1: Gantt Chart of Dog Adoption System

Chapter 4: High Level Design of System

4.0.1 Data Modeling: ER Diagram

This ER diagram shows a dog adoption system with three main entities: Users, Dogs, and Admin. Users have an ID, name, email, and password, and they can submit adoption forms for dogs. Dogs have a unique Dog_ID, breed, image, and age. Admins have an ID, name, email, and password, they manage the records of dogs and can receive multiple adoption requests from users.

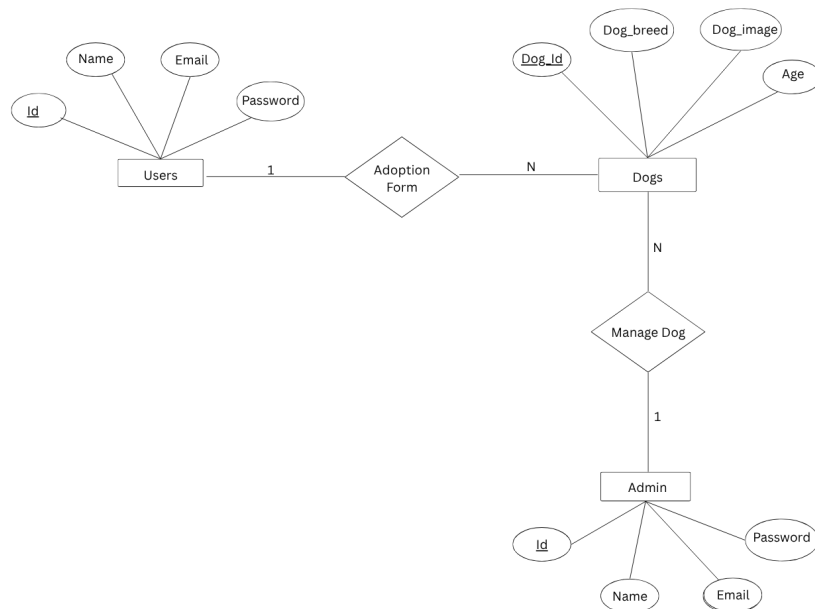


Figure 4.1: ER Diagram of Dog Adoption System

4.0.2 Process Modeling: DFD

This diagram shows the interaction between users, admins, and the dog adoption system. Users can create accounts, view available dogs, and fill out adoption forms. Admins can add or delete dog records, view adoption applications, and accept or reject them. Both users and admins connect to the dog adoption system to perform their respective tasks. It uses processes, data stores and also flow of data.

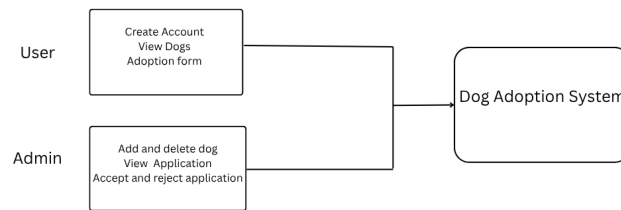


Figure 4.2: DFD Level 0 of Dog Adoption System

The diagram illustrates a streamlined pet adoption platform where user can browse available dogs (displaying breed, age, and an image) on a user homepage. To proceed with an adoption, a guest must register with their name, email, and password, then log in to become a verified user. Verified users access their dashboard to complete an adoption form providing details like their full name, email, preferred dog, and address which is submitted to the admin. Administrators, who also log in separately, can create or delete dog entries and review submitted adoption forms, choosing to approve or reject them.

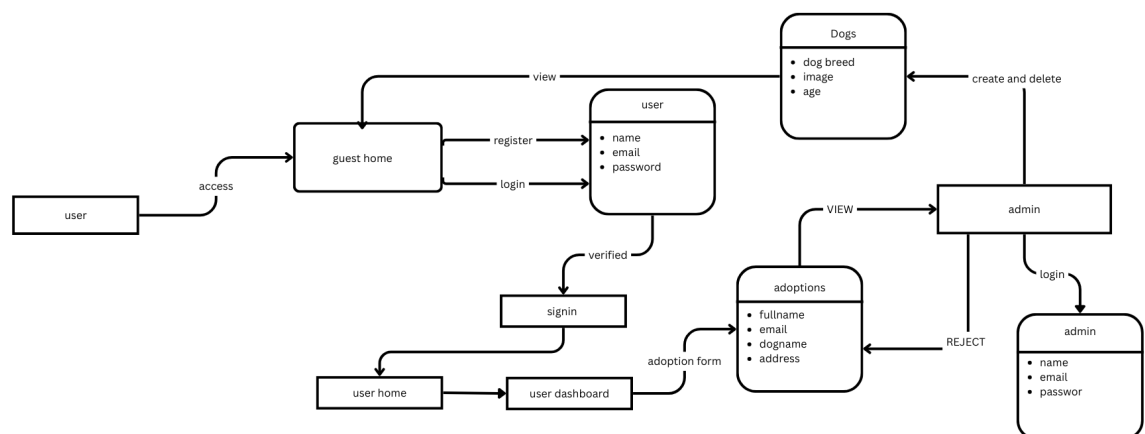


Figure 4.3: DFD Level 1 of Dog Adoption System

4.1 System Design

The use case diagram of dog adoption highlights the different roles and their interactions with the system. This use case diagram shows a dog adoption system where users can view available dogs and fill out adoption forms, while admins can add new dogs, view submitted forms, and either accept or reject adoption requests.

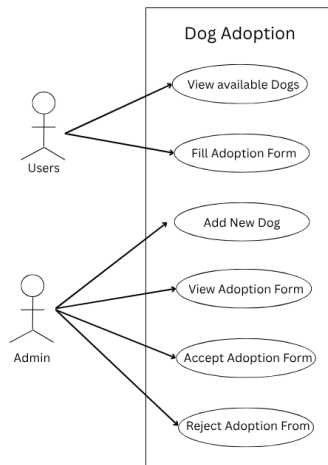


Figure 4.4: Use-Case Diagram of Dog Adoption System

4.1.1 Architectural Design

This diagram illustrates how both users and admins interact with a web application. They access it through clients (web browsers), where the frontend built with HTML, CSS, and JavaScript handles the user interface. The frontend communicates with the backend (PHP), which processes requests and interacts with the database to store or retrieve information.

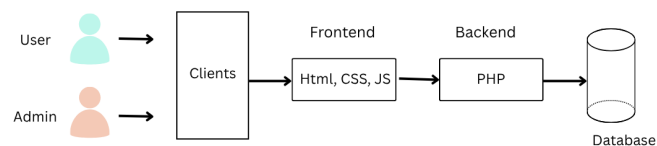


Figure 4.5: Architectural Design of Dog Adoption System

4.1.2 Database Schema Design

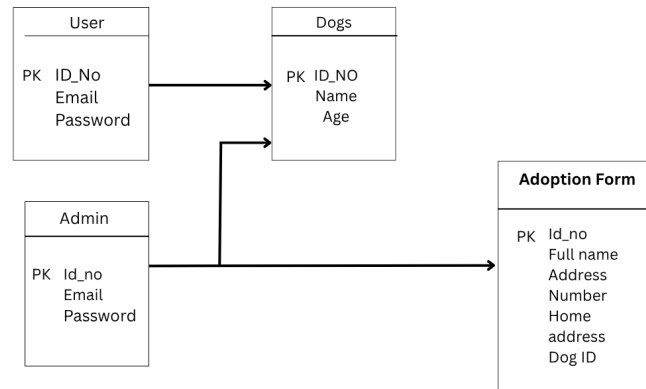


Figure 4.6: Database Schema of Dog Adoption System

4.1.3 Interface Design (UI/UX)

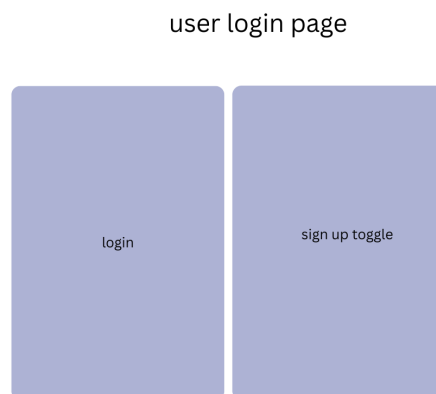


Figure 4.7: Log In and Sign Up of Dog Adoption System

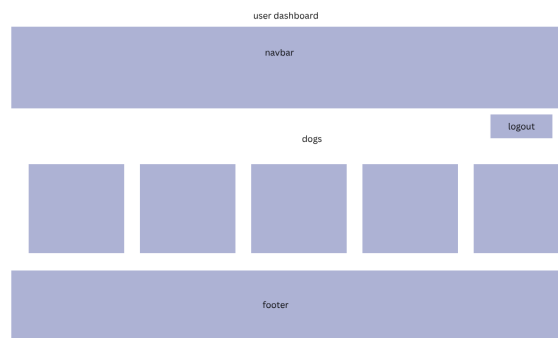


Figure 4.8: User Dashboard of Dog Adoption System

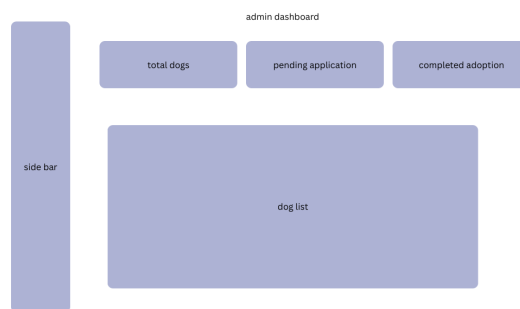


Figure 4.9: Admin Dashboard of Dog Adoption System

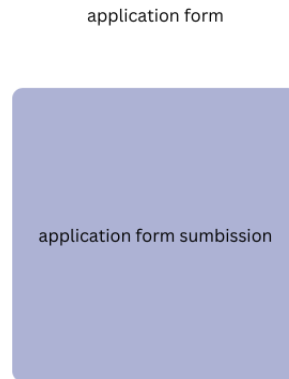


Figure 4.10: Adoption Form Submission of Dog Adoption System

4.1.4 Physical DFD

The diagram outlines a streamlined pet adoption platform where users interact via a browser to view dog listings (including breed, age, image) and navigate from a home page to a dashboard. From there, they complete an adoption form (with fields like full name, email, number, and address) which generates application data. That data can be viewed, accepted, or rejected by an admin, who logs in using their credentials. Admins also have the ability to add or delete dog listings.

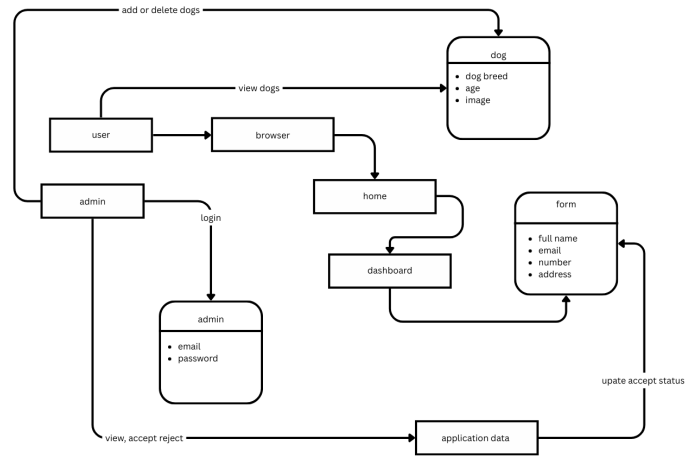


Figure 4.11: Physical DFD of Dog Adoption System

Chapter 5: IMPLEMENTATION AND TESTING

5.1 Implementation

5.1.1 Tools Used

The Dog Adoption System was developed using a combination of web development technologies to ensure an interactive, responsive, and functional platform. HTML (HyperText Markup Language) was utilized to structure the content and layout of the webpages, providing the foundation for the system's interface. CSS (Cascading Style Sheets) was employed to enhance the visual presentation, including the design, colors, fonts, and overall aesthetics, ensuring a user-friendly and visually appealing experience. PHP (Hypertext Preprocessor) was used as the server-side scripting language to handle dynamic content, process user requests, manage adoption forms, and facilitate communication between the application and the database. Together, these tools provided an efficient, scalable, and maintainable framework for implementing the system's core functionalities.

5.2 Testing

The process of checking the quality, functionality, and performance of a project before deployment is called testing. It is necessary to test the project to find out whether it is working as per the expectation of the client or not. It is done to find if there is any error in the site before launching it.

5.2.1 Test Cases for Unit Testing

Table 5.1: Testing Table for Admin

Test ID	Test Case	Status	Remarks
ID-1	Login	Pass	Works fine
ID-2	Invalid Password	Pass	Invalid Password
ID-3	Logout	Pass	Redirects to homepage
ID-4	Add Dog	Pass	Successfully Added
ID-5	Delete Dog	Pass	Removed Successfully
ID-6	Application Form	Pass	Accept and Rejected

Table 5.2: Testing Table for User

Test ID	Test Case	Status	Remarks
ID-1	Login	Pass	Works fine
ID-2	Invalid Password	Pass	Invalid Password
ID-3	Logout	Pass	Redirects to homepage
ID-4	View Dog	Pass	Redirected to dog list
ID-5	Application Form	Pass	Accept and Rejected

5.2.2 Test Cases for System Testing

Table 5.3: Testing Table for system testing

Test ID	Test Case	Status	Remarks
ID-1	Login	Pass	Redirect to dashboard
ID-2	Add dog	Pass	Update dog table
ID-3	Delete dog	Pass	Removes from table
ID-4	Application form	Pass	Approve and delete

Chapter 6: CONCLUSION AND FUTURE RECOMMENDATION

6.1 Conclusion

In conclusion, the dog adoption system serves as a comprehensive and efficient platform that facilitates the entire adoption process through a user-centric approach. By enabling users to create secure accounts, the system ensures personalized access, allowing potential adopters to submit detailed adoption forms with ease and accuracy. This digital framework not only streamlines the administrative workflow for managing adoption requests but also promotes transparency and accountability throughout the process.

Moreover, the system enhances communication between the shelter and prospective adopters, ensuring that all necessary information is collected and verified before finalizing adoptions. By integrating user authentication and structured application submission, the platform supports responsible adoption practices and helps safeguard the welfare of the dogs.

6.2 Lesson Learnt

Throughout the development and implementation of the dog adoption system, several valuable lessons were learned that highlight both the challenges and opportunities in creating an effective digital adoption platform. While working on the dog adoption system, we learned several important things. First, it is very important to make the website easy to use so that anyone can create an account and fill out the adoption form without confusion. Clear communication with users helps keep them interested and confident during the adoption process. We also realized how important it is to keep user information safe and organized to avoid problems.

Listening to feedback from both adopters and shelter staff helped us understand what works well and what needs improvement. We saw how technology can make it easier for more dogs to find good homes. Finally, we learned that the system needs regular updates to stay useful and meet the needs of both users and the shelter.

6.3 Future Recommendation

To further improve the dog adoption system, several enhancements are recommended for future development. One significant improvement would be the creation of a dedicated mobile application for both Android and iOS devices, which would allow users to access the platform conveniently from their smartphones or tablets. Additionally, integrating real-time chat support would enhance communication between potential adopters and shelter staff, providing immediate assistance and clarifications throughout the adoption process. The implementation of advanced search and filtering options based on criteria such as breed, age, size, and temperament would enable users to find dogs that best suit their preferences more efficiently.

Furthermore, the system could benefit from automated notifications and reminders via email or SMS, keeping users updated on application statuses, upcoming appointments, and vaccination schedules. Incorporating detailed health and behavioral records for each dog would provide adopters with comprehensive information to make informed decisions. To increase the visibility of available dogs, social media integration should be considered, allowing easy sharing and promotion of dogs in need of homes. Expanding the platform to include volunteer and foster home management modules could also support shelter operations by organizing additional resources and care for the animals.

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Chapter 8: Appendices

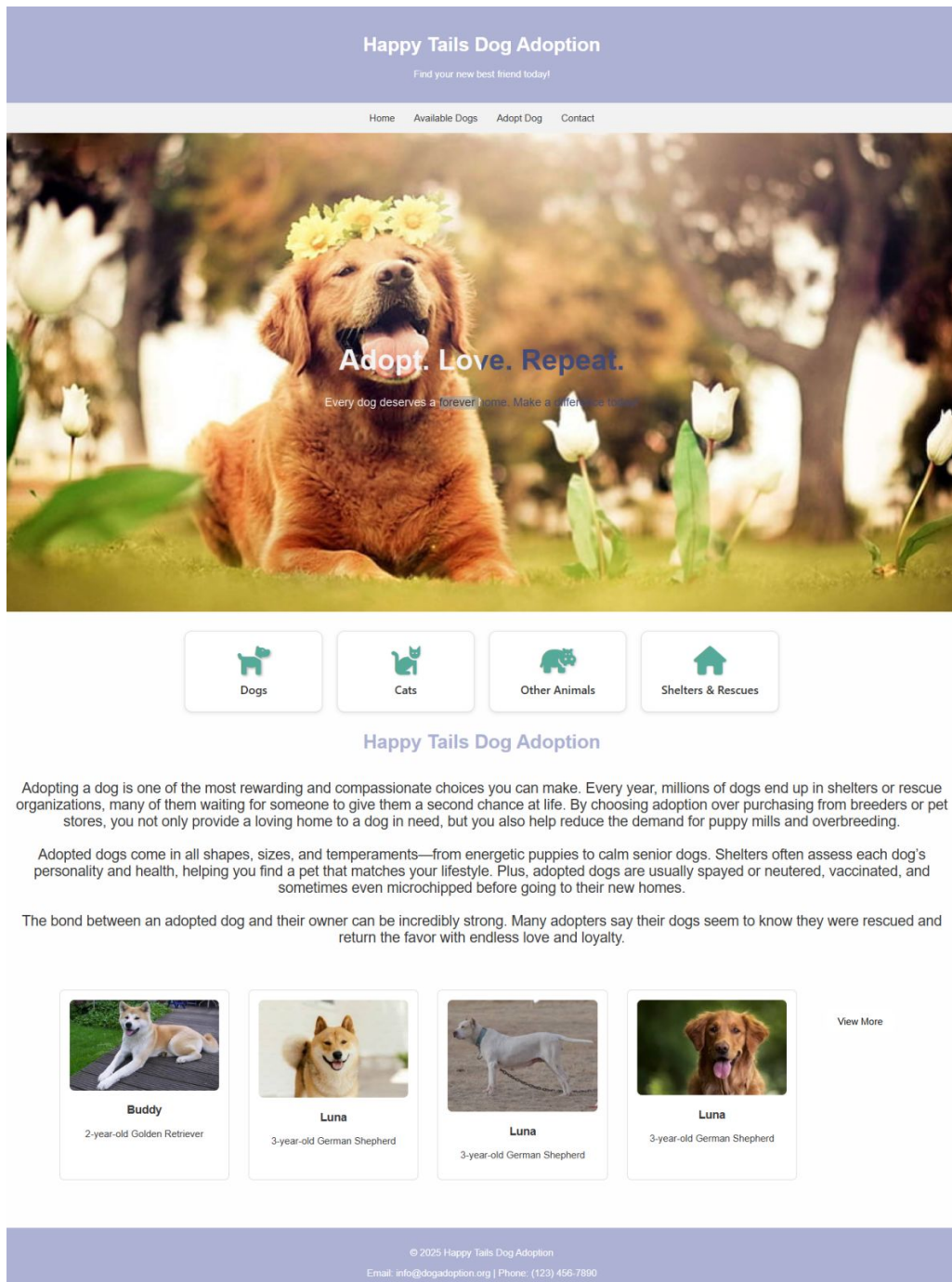


Figure 8.1: Home Page of Dog Adoption System

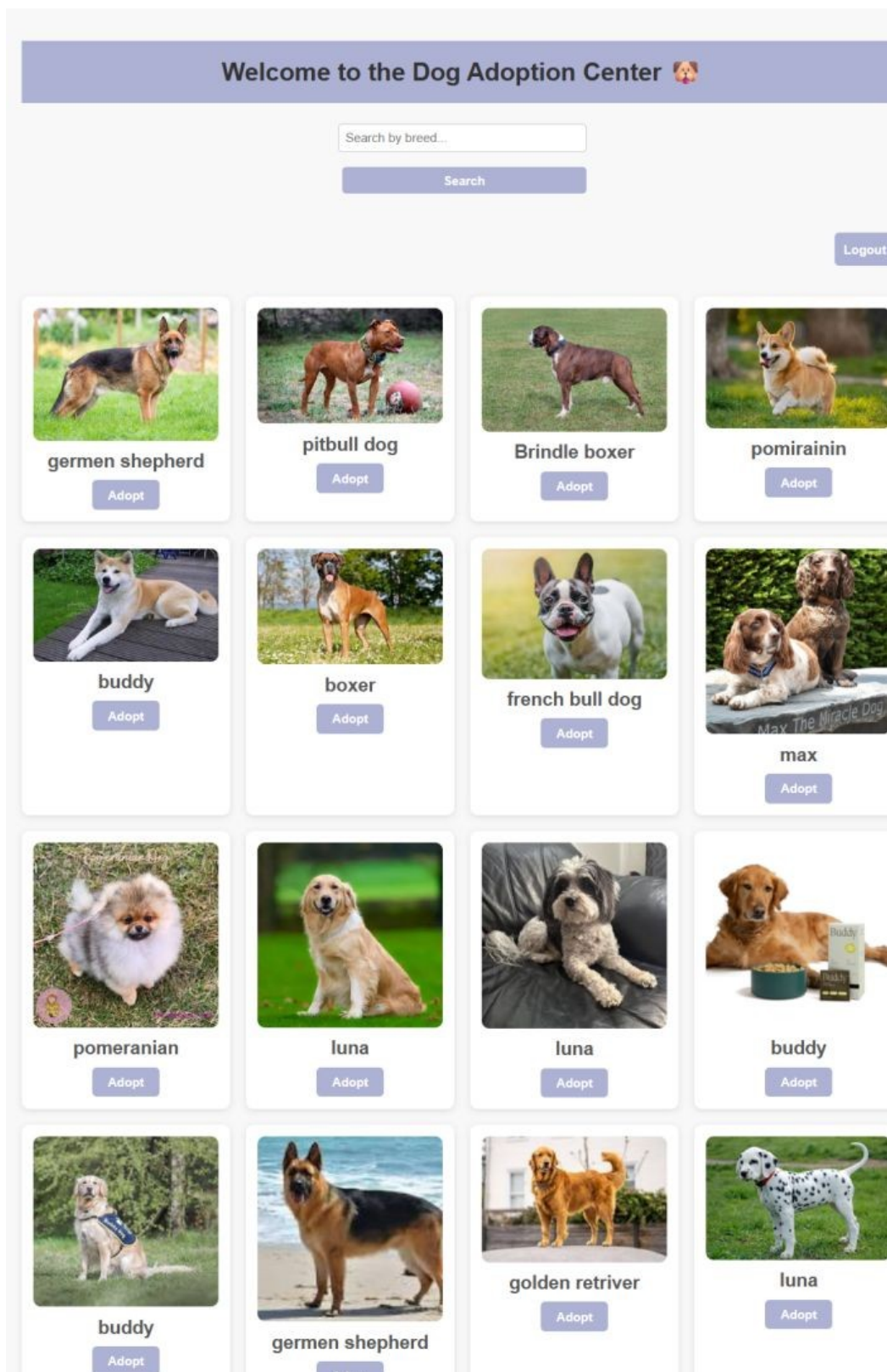


Figure 8.2: User Dashboard of Dog Adoption System

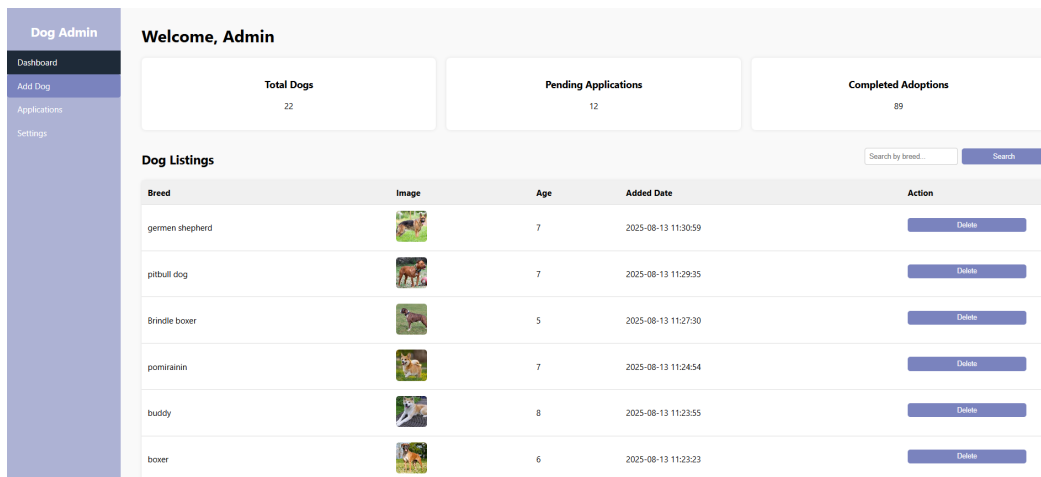


Figure 8.3: Admin Panel of Dog Adoption System

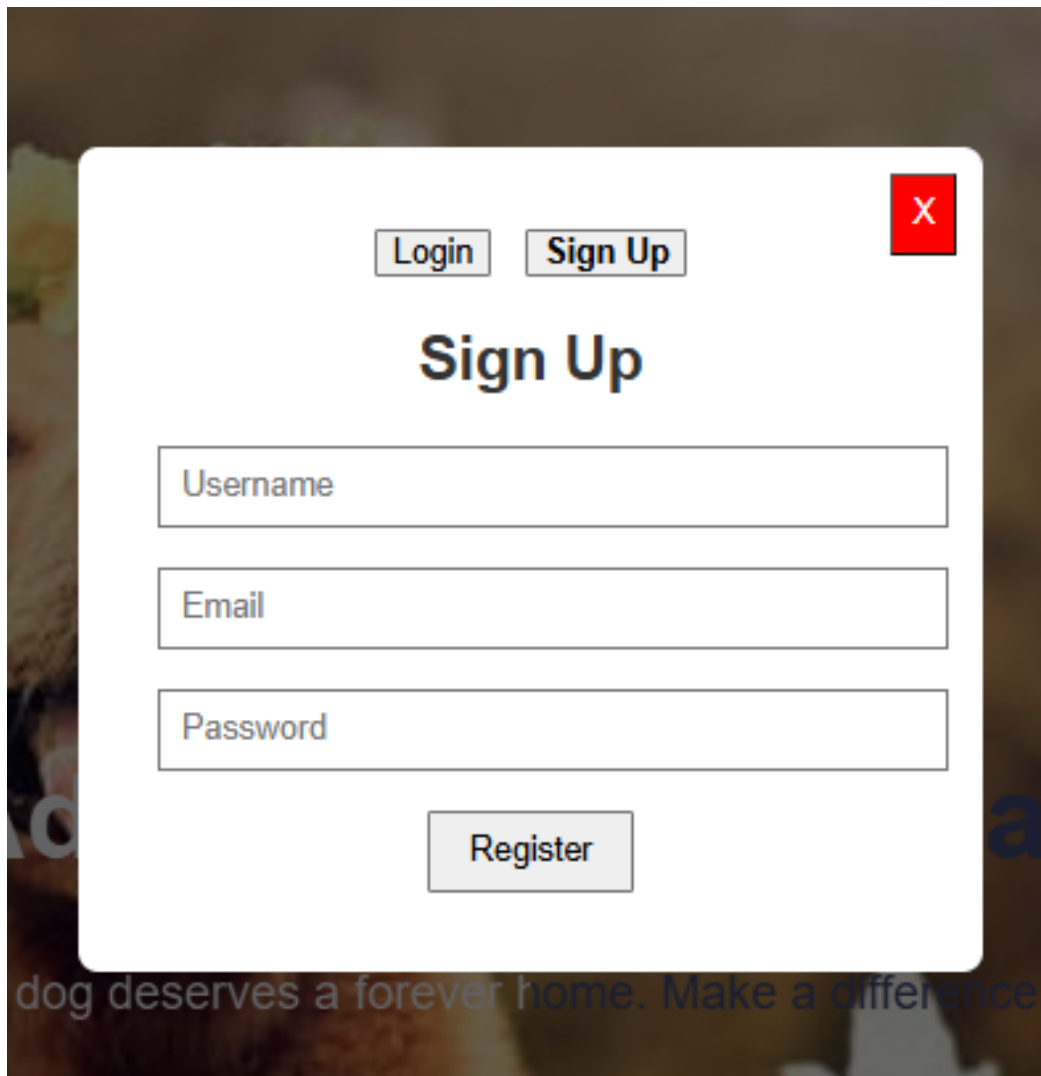
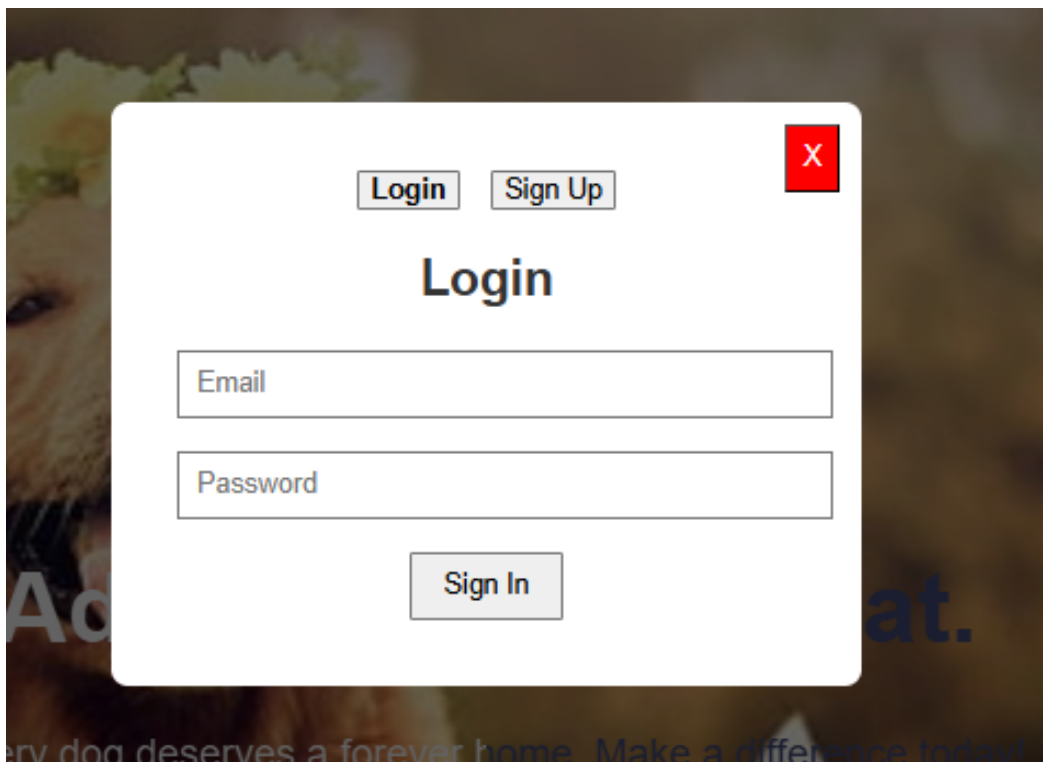


Figure 8.4: Sign Up Form of Dog Adoption System



The image shows a login modal window for a dog adoption system. The modal is white with rounded corners and a red close button (X) in the top right corner. At the top, there are two buttons: "Login" and "Sign Up". Below these is the title "Login" in a large, bold, blue font. There are two input fields: "Email" and "Password". Below the input fields is a "Sign In" button. The background of the page is a blurred image of a dog's face.

Login Sign Up

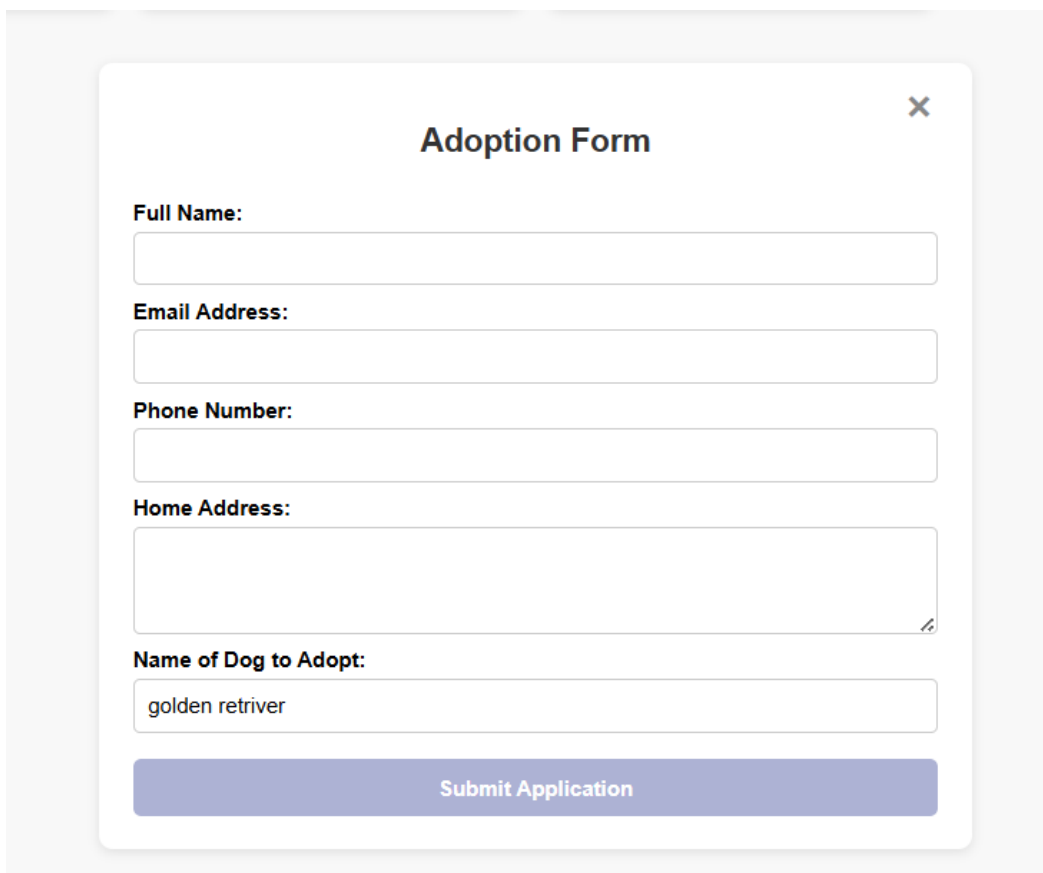
Login

Email

Password

Sign In

Figure 8.5: Log In Page of Dog Adoption System



The image shows an adoption form modal window. The modal is white with rounded corners and a close button (X) in the top right corner. The title "Adoption Form" is centered at the top. Below the title are five input fields with labels: "Full Name:", "Email Address:", "Phone Number:", "Home Address:", and "Name of Dog to Adopt:". The "Name of Dog to Adopt:" field contains the text "golden retriever". At the bottom of the form is a blue button labeled "Submit Application".

Adoption Form

Full Name:

Email Address:

Phone Number:

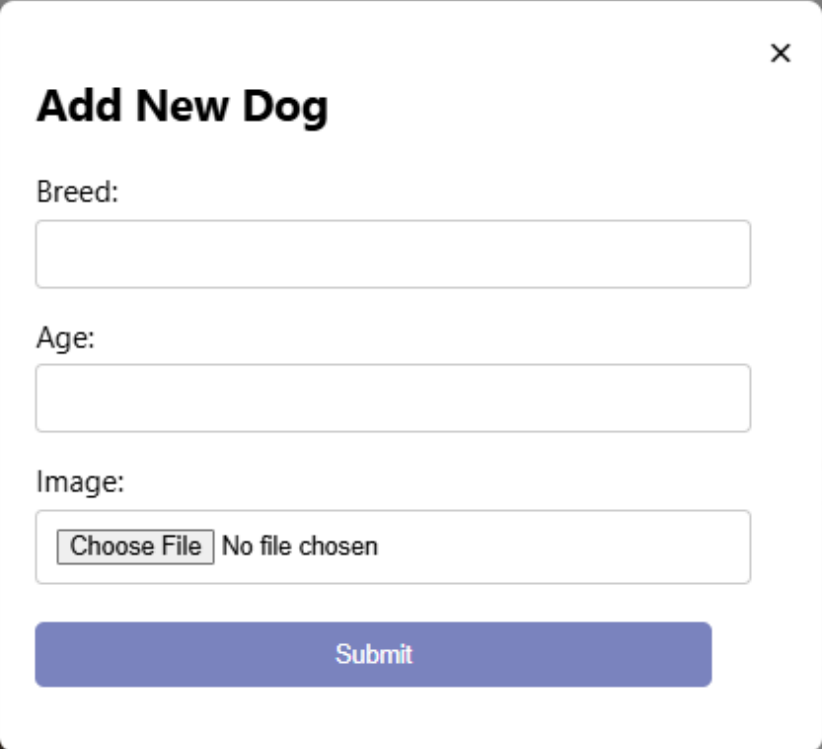
Home Address:

Name of Dog to Adopt:

golden retriever

Submit Application

Figure 8.6: Adoption Form of Dog Adoption System



The image shows a modal form titled "Add New Dog" with a close button (X) in the top right corner. The form contains three input fields: "Breed:", "Age:", and "Image:". The "Image:" field includes a "Choose File" button and the text "No file chosen". A blue "Submit" button is at the bottom. The background is a blurred view of a dog adoption system interface with columns for "Image", "Name", and "Adopted Date".

Image	Name	Adopted Date
		2025-08-13 09:00
		2025-08-13 09:00

Figure 8.7: Add Dog Form Dog Adoption System